QUIZ #3 Sec 1

Consider the ODE:

$$(x-1)(x+1)^2 y'' + x \cdot y' - 2y = 0$$

Let $\varphi_{\pm} = (x \mp 1)^r \sum_{k=0}^{\infty} a_k (x \mp 1)^k$

solve the ODE around (but not necessarily at) the singular points $x = \pm 1$.

Does $\lim_{x \to \pm 1} \varphi_{\pm}$ exist?

Solution

- Observe

$$\lim_{x \to -1} (x+1)\frac{Q}{P} = \lim_{x \to -1} \frac{x}{x^2-1}$$ does not exist

Therefore $x=-1$ is not a regular singular point. Frobenius Thm says nothing about φ_-.

$$\lim_{x\to 1}(x-1)\frac{Q}{P} = \frac{1}{4} = a_0$$

& $$\lim_{x\to 1}(x-1)^2\frac{R}{P} = 0 = b_0$$

So $x=+1$ is a regular singular point. By Frob. Thm, φ_+'s bad behavior at $+1$ (if any) comes from x^r. The indicial eqn reads:

$$r(r-1)+r/4+0=0$$

$$\Rightarrow r_2=0,\ r_1=3/4$$

Since $r_1 - r_2 \notin \mathbb{Z}$,

for each root $r_{1,2}$,

$$a_{1,2} = (x-1)^{r_{1,2}} \sum_{k=0}^{\infty} a_k (x-1)^k$$

is a solution and

$$\lim_{x \to 1} a_{1,2} = a_0 \cdot \lim_{x \to 1} (x-1)^{r_{1,2}} = \begin{cases} a_0, & r_2 \\ 0, & r_1 \end{cases}$$

for